BRANCH-AND-BOUND

- Branch-and-bound is a general technique for improving the searching process by systematically enumerating all candidate solutions and disposing of obviously impossible solutions.
- It involves a bounding function.
- Systematic enumeration of all candidate solutions, discarding large subsets of fruitless candidates by using upper and lower estimated bounds of quantity being optimized.
- It is used to solve optimization problem.
- Branch-and-bound usually applies to those problems that have finite solutions, in which the solutions can be represented as a sequence of options.
- The first part of branch-and-bound branching requires several choices to be made so that the choices will branch out into the solution space.
- In these methods, the solution space is organized as a tree like structure. (State-Space tree)
- The term branch-and-bound refers to all state space search methods in which all children of the E-node are generated before any other live node can become the E-node(BFS)
- It completely searches the stat-space tree to get optimal solution.

Terminology

- Definition 1 Live node is a node that has been generated but whose children have not yet been generated.
- Definition 2 E-node is a live node whose children are currently being explored. In other words, an E-node is a node currently being expanded.

- Definition 3 Dead node is a generated node that is not to be expanded or explored any further. All children of a dead node have already been expanded.
- Definition 4 Branch-and-bound refers to all state space search methods in which all children of an E-node are generated before any other live node can become the E-node.

BRANCH AND BOUND STRATEGIES

- A Branch-And-Bound algorithm consists of a systematic enumeration of candidate solutions by means of state space search: the set of candidate solutions is thought of as forming a rooted tree with the full set at the root.
- The algorithm explores branches of this tree, which represents subsets of the solution set.
- Before enumerating the candidate solutions of a branch, the branch is checked against upper and lower estimated bounds on the optimal solution, and is discarded if it cannot produce a better solution than the best one found so far by the algorithm.
- It may traverse the tree in any following manner:

1. **BFS(Breadth First Search) or FIFO Branch and Bound**
2. **D-search or LIFO Branch and Bound**
3. **Best First Search or Least Cost Search**

BFS is an FIFO search in terms of live nodes

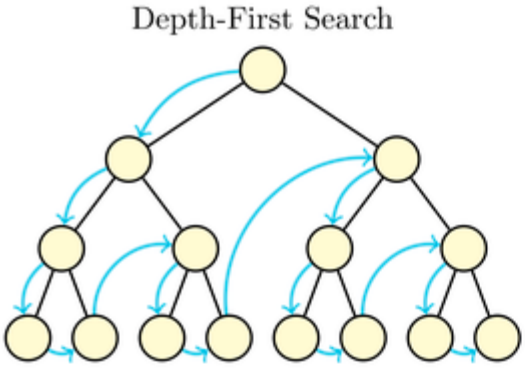
- In branch-and-bound terminology, a BFS-like state space search will be called FIFO (First In First Out) search as the

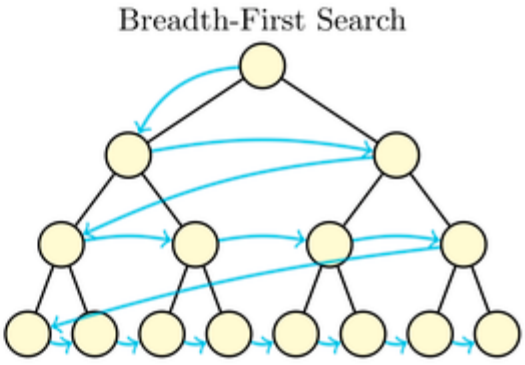
list of live nodes is a first-in-first-out list (or queue).

- List of live nodes is a queue
  - ✓ FIFO (First In, First Out): Always the oldest node in the queue is used to extend the branch.
  - ✓ This leads to a breadth-first search, where all nodes at depth d are visited first, before any nodes at depth d+1 are visited.

DFS is an LIFO search in terms of live nodes

- A D-search-like state space search will be called LIFO (Last In First Out) search as the list of live nodes is a last-in-first-out list (or stack).
- List of live nodes is a stack
  - ✓ LIFO(Last In First Out):Always the youngest node in the stack is used to extend the branch
  - ✓ This leads to Depth-First Search, where the branch is extended through every 1st child discovered at a certain depth ,until a leaf node is reached





Best First Branch and Bound

- Improve breadth-first strategy.

- After visiting all children of a given node, we look at all promising and unexpanded nodes.
- Expand the node with the best bound first.
- Implemented using a priority queue.